MILITARY SPECIFICATION

FUEL SERVICING UNIT, AIRCRAFT, TRAILER MOUNTED, 600 GPM, AIR TRANSPORTABLE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trailer-mounted unit used for transferring and filtering hydrocarbon fuels for direct aircraft fueling and defueling operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-440	- Clips, Electrical, General Specification For.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.

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MIL P-116	- Preservation-Packaging, Method of.
MIL-B-121	- Barrier Material, Grease Proofed, Waterproofing, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical, Packaging, of.
MIL-G-5572 - Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-F-8901 - Filter-Separators, Liquid Fuel: and Filter-Coalescer Elements, Fluid Pressure: Inspection Requirements and Procedures for.
MIL-E-10062 - Engines: Preparation for Shipment and Storage.
MIL-F-52308 - Filter Element, Fluid Pressure.
MIL-M-81380 - Monitor, Contamination, Fuel Dispensing System.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-810 - Environmental Test Methods.
MS27023 - Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Flanged, Type IV.
MS27027 - Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Flanged, Type VII.
MS27030 - Gasket, Coupling Half, Quick Disconnect, Cam-Locking Type.
MS51335 - Pintle Assembly, Towing, 18,000 pounds Capacity, Manual Release.
MS51336 - Lunette-Coupler, Drawbar, Ring.
MS75021 - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

DRAWINGS

ARMY MOBILITY EQUIPMENT COMMAND

D13216E2773 - Canister.
D13216E2774 - Washer, Spring Tension.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Motor Carrier Safety Regulations.

NATIONAL BUREAU OF STANDARDS (NBS)

H44 - Specifications, Tolerances, and Regulations for Commerical Weighing and Measuring Devices.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Specification for Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet, Sheet, and Strip.
A514 - Specification for High-Yield-Strength, Quenched and Tempered Alloy Steel Plate, Suitable for Welding.
A517 - Specification for High-Strength Alloy Steel Plates, Quenched and Tempered, for Pressure Vessels.
B209 - Specification for Aluminum and Aluminum-Alloy, Sheet and Plate.
B211 - Specification for Aluminum Alloy Bars, Rods, and Wire.
B308 - Specification for Aluminum-Alloy Standard Structural Shapes, Rolled or Extruded.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The fuel servicing unit shall be trailer mounted, air transportable, and shall be capable of transferring and filtering JP-4 and JP-5 jet fuels and aviation gasoline at the pressures, flow rates, and purity specified herein. The components shall be assembled and mounted on a trailer frame having four retractable spring-mounted wheels. The entire unit shall have a width not to exceed 7 feet 4 inches, an overall length not to exceed 13 feet, an overall height not to exceed 5 feet 9 inches with width in towing position and 5 feet with wheels retracted. The fuel servicing unit shall

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include the following major components: A liquid-cooled diesel engine close coupled to a pump; an automatic pressure-flow rate controlling fueling valve; fuel-defuel valving; filter-separator case with elements, segregator canisters, automatic air eliminator, automatic and manual water drain with sump heater; liquid flow meter; line strainer; contamination monitor with elements; thermal, pressure, and fuel shutoff safety controls; a purging system to purify the unit and discharge hoses; a drain down system; and a bonded grounding system.

3.2 First article. When specified (see 6.2), the contractor shall furnish a fuel servicing unit for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Performance.

3.3.1 Environmental conditions. The fuel servicing unit shall be capable of withstanding the following:

3.3.1.1 Storage temperatures. The fuel servicing unit shall not be damaged by storage in ambient temperatures ranging from -40° degree Fahrenheit (°F) to +160°F for not less than 50 hours.

3.3.1.2 Operating temperatures. The fuel servicing unit shall be capable of performing, as specified herein, in ambient temperatures between -25°F and +125°F.

3.3.2 Pumping system operations. The filter-separator, filter elements, and canisters shall be capable of removing free water and solid contaminates from the fuels in accordance with the tests, performance, and pressures requirements of MIL-F-8901, and at the flow rates and operating pressures specified herein. The entire unit shall be designed to provide protection against explosion or fire while operating under fueling, defueling, or standby idle speed conditions. When a change from flow to shutoff occurs at the nozzle or discharge end of a fueling hose, the engine shall automatically and immediately decelerate to idle speed and continue at idle speed, without manual attention, and automatically accelerate to the preset pumping speed when the discharge is again opened. Fuel flow shall be in one direction only through the pump, filter-separator, fueling valve, contamination monitor, and meter. Automatic shutdown of the unit shall occur when the differential pressure across the contamination monitor, or the filter-separator elements exceeds a predetermined level; when the water level in the filter-separator sump exceeds a normal level; or when the fuel temperature in the pump volute exceeds 135°F ±5°F. All fuel handling components shall be designed for a working pressure of 150 pounds-force per square inch (psi) and shall be capable of withstanding a hydrostatic pressure of 225 psi. Suitable devices shall be incorporated to prevent surge or other dynamic pressures exceeding 120 psi, in the unit and discharge hose, at a nozzle closure speed of 1/2 second when fueling at a rate of 600 gallons per minute (gpm) through 75 feet of 4-diameter fueling hose. The fuel servicing unit shall have the performance capability for the following normal functions, and shall exhibit no instability in pressure, flow rates, or engine operation for any function:

a. Fueling with jet fuels at rates of 300 and 600 gpm while maintaining a pressure of 50 psi, ±5 psi, at the discharge piping manifold, and a vacuum of not less than 8 inches of mercury at the suction, while

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discharging through 75 feet of 4-inch diameter fueling hose through a center point nozzle.

b. Fueling with aviation gasoline at rates of 150 gpm through one 2-inch discharge hose and 300 gpm through two hoses, while maintaining a pressure of 30 psi at the discharge piping manifold through overwing nozzle(s).

c. Defueling of jet fuels and aviation gasoline from aircraft at a rate of 150 gpm through a 4-inch diameter fuel-defuel hose at a controlled vacuum of 8 inches of mercury; and shall also provide override for collapse and evacuation of the hose.

The fueling servicing unit system shall provide the performance specified herein when handling fuels conforming to MIL-G-5572 and MIL-T-5624.

3.4 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual componenets or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5.1 Steel. Rolled steel shapes and plates shall be high strength alloy steel conforming to ASTM A514 or A517, grade F.

3.5.2 Corrosion-resisting steel. Corrosion-resisting steel shall conform to ASTM A167, type 304.

3.5.3 Aluminum alloy. Aluminum for plates and sheets shall conform to ASTM B209, alloy 5083 or alloy 5086. Aluminum for rods and bars shall conform to ASTM B211, alloy 6061. Aluminum for structural shapes shall conform to ASTM B308, alloy 6061. The temper of the aluminum alloy used shall be appropriate for the end use.

3.5.4 Gaskets and seals. Gaskets and seals shall be suitable for use with hydrocarbon fuels.

3.6 Components. The fully equipped fuel servicing unit shall have an overall width of not more than 88 inches, an overall length of not more than 13 feet, and an overall height of not more than 69 inches with wheels in towing position and 60 inches with wheels retracted.

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3.6.1 Trailer. The trailer shall be designed to support and transport the components specified herein when in a drained condition under the mobility and air transportability requirements specified herein. The trailer, with wheels down, shall be capable of negotiating aircraft ramps having an approach angle of +20° from level, and the rear of chassis shall be provided with rollers if required to prevent gouging of ramp or damage to the chassis. The trailer shall be designed for traversing highways at speeds of 45 miles per hour (mph), and rough terrain at speeds up to 25 mph. The trailer shall also support, during air transport, a 6,000 pound (lb) transporter trailer utilizing the stanchions as specified in 3.6.3.

3.6.1.1 Frame. The frame shall be constructed with sufficient strength to withstand all the loads imposed during the mobility and air transportability operating requirements specified herein. The frame shall be fabricated of high strength alloy steel or aluminum alloy.

3.6.1.2 Wheels. Four wheels shall be provided of retractable non-steering, spring mounted design. Manual hydraulic pumps and cylinders shall be provided, for raising and lowering of wheels. Not less than one manual hydraulic pump and two cylinders shall be provided for each pair of wheels.

3.6.1.3 Tires. The tires shall be capable of withstanding all the loads imposed specified herein.

3.6.1.4 Tongue. A removable tongue shall be provided on the front, with a lunette ring conforming to MS51336. The tongue shall have a hinged connection permitting lowering and raising and a pin for securing in vertical position.

3.6.1.5 Towing pintle. A towing pintle assembly, 18,000 lb capacity, conforming to MS51335, Part No. MS51335-2 shall be attached to the rear frame crossmember.

3.6.1.6 Lighting system. The lighting system shall be manufacturer's standard 12-volt (V) direct current potential and shall comply with DoT Motor Carrier Safety Regulations, sections 393.14, 393.20, 393.25 through 393.29, 393.32 and 393.33. The systems shall include at least two spotlights and tailights, directional signals, front and rear marker and clearance lights, and reflector devices. Lights and reflectors shall be recesed or guarded to protect them from damage during normal operation of the trailer. All wiring shall be properly routed and anchored to protect against snagging or tearing away.

* 3.6.1.7 Receptacle system, 24 volt. The trailer shall be equipped wth one 12-contract receptacle with cover, and one 7-contact receptacle with cover, installed at the front of the trailer. The 12-contact receptacle shall conform to MS75021, Part No. MS75021-1. The 7-contact receptacle shall conform to SAE J560 round socket for jumper cable. The 12-contact receptable shall be provided with resistance to each circuit to reduce the voltage of a tactical military design towing vehicle from 24V to 12V. The 7-contact socket shall be wired in accordance with SAE J560 to supply 12V directly to the trailer circuits. The 24V, 12-contact receptacle shall be connected through the necessary resistors into the 12V trailer circuits as follows:

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A - Do not connect.
B - Connect to yellow circuit.
C - Do not connect.
D - Connect to white circuit.
E - Connect to black and brown circuits.
F - Do not connect.
H - Do not connect.
J - Connect to green circuit.
K - Connect to blue circuit (if auxiliary circuit is used in trailer wiring).
L - Connect to white circuit.
M - Do not connect.
N - Do not connect.

Circuits B and J on tactical truck tractors are combination stop and turn indicator circuits. The normal 12V turn signal lights will function both as turn signals and stoplights and the normal 12V stoplights will not be in operation when the trailer is connected to a towing vehicle with a 24V power supply. Because of this condition, the stoplight red circuit is not connected to the 24V, 12-contact receptacle.

3.6.2 Lifting and tiedown attachments. Not less than four attachments shall be provided to permit lifting and tiedown of the unit in its normal position. The attachments shall permit tiedown of the unit to the floor or deck of the transportation medium. The attachments shall be of such design, location, and number as to enable the unit to be anchored in the aircraft to prevent shifting or movement in any direction, under the loading conditions specified herein. The attachments shall be located so as to permit easy connection of cables or turnbuckles. Attachments shall be indicated by a suitable information plate, by stenciling, or other suitable marking of the unit. The marking shall clearly indicate that the device is intended for tiedown of the unit on the carrier when shipped.

3.6.2.1 Flight and taxiing loads. The equipment shall be designed to withstand, without loss of serviceability, an acceleration of 3 gravity (g) for a minimum of 0.1 second applied dynamically and independently along each of the longitudinal and vertical axis in each direction, and 1.5g for a minimum of 0.1 second applied dynamically and independently along the laterial axis in each direction.

3.6.2.2 Emergency landing loads. Equipment shall be designed to withstand the following loads encountered in crash landings, without any of the major components of items being transported breaking loose. The item need not be serviceable after being subjected to such accelerations.

a. A minimum of 8g in either direction applied independently along each horizontal axis for a minimum of 0.1 second. When the equipment is of such size or configuration that it can be loaded only into cargo aircraft with a particular axis parallel to the longitudinal axis, the 8g requirement need be met only along this axis and in only the forward direction when the loading position is fixed or specified.
b. A minimum of 4-1/2g vertically downward for a minimum of 0.1 second in such a direction that equipment carried, or in a cargo compartment, imposes a load on its wheels or supports in a downward direction.

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3.6.3 Stanchions. Four stanchions, fabricated from aluminum alloy 6062-T6, or high strength alloy steel, square or round tube, shapes, and plates, shall be attached to support a total design static load of 6,000 lb. The stanchions, so loaded, shall withstand the vertical load indicated in 3.6.2.2. The stanchions shall be designed to accommodate and support a transporter trailer shown on Figure 1. The top of each stanchion shall have a cradle to support and lock around the axle of the transporter trailer as shown on Figure 2.

3.6.4 Stowage racks or boxes. Stowage racks or expanded metal boxes shall be provided of suitable capacity to store the following:

a. Two 75-foot lengths of 2-inch fuel hose.
b. Two 75-foot lengths of 4-inch fuel hose.
c. Overwing and center point nozzels.
d. Six spare canisters (see 3.6.7.3).
e. Sixty spare filter-separator elements (see 3.6.7.4).
f. Eighty space contamination monitor elements (see 3.6.8).
g. Two ground rods (see 3.6.15).
h. Spare parts and tools (see 3.6.16).

The hoses and nozzles listed in a, b, and c are not to be furnished. The stowage devices shall be detachable, shall be securing pins with attachment chains, and shall not interfere with the piggyback carry of a trailer shown in Figure 1.

3.6.5 Pump. Pump shall be self-priming, centrifugal type, close coupled to the diesel engine. Pump shall be of standard commercial manufacture of aluminum utilizing replaceable wear and corrosion-resisting parts and self-lubricating seals and bearings. Pump inspection and cleanout shall be without removal of suction or discharge connections. Pump, seals, and impeller shall be suitable for the fuels, pressures, and flow rates specified herein. Pump discharge pressure shall not exceed 120 pounds-force per square inch (psi) at maximum recommended engine revolutions per minute and no flow condition.

3.6.6 Engine. The pump shall be powered by a multiple cylinder, liquid cooled, diesel engine. The engine shall be capable of satisfactorily performing utilizing grade JP-4 or JP-5 fuel in accordance with MIL-T-5624 or diesel fuel. The power and speed required of the engine to meet the performance requirements of the pump shall not exceed the continuous horsepower rating at the applicable speed. The engine shall be furnished complete with accessories, and shall include at least the following:

a. A fuel tank with capacity of 8 hours normal operation. The fuel tank shall be equipped with a float valve or fuel lines arranged for accepting JP-4 or JP-5 fuel immediately downstream from the filter-separator (see 3.6.6.2). When unit is used for dispensing aviation gasoline, fuel tank with diesel fuel must be used.
b. A spark and flame arresting muffler directed away from the operator.
c. An overspeed shutdown control.
d. Controls to automatically shut off the engine due to high coolant temperature and low lubricating oil pressure.
e. A control to automatically shut off the engine when the temperature

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of the fuel being pumped rises to 135°F in the pump volute.
f. A lubricating oil pressure gage.
g. A cooling liquid temperature indicator.
h. A tachometer.
i. An hour meter.
j. A hydraulic cranking system.
k. A remote control engine governor having not less than three speed positions.
l. A manual engine control, of the push-pull, locking micrometer type.
m. A removable engine housing.

3.6.6.1 Mounting. The engine and pump shall be mounted in such a manner as to assure perfect pump and engine alinement during operation and towing over tough terrain. Isolation of engine and pump vibrations from the chassis and other components shall be provided.

3.6.6.2 Engine fuel source. The manual fueling valve for the fuel tank shall be connected downstream from the filter-separator.

3.6.6.3 Hydraulic cranking system. The hydraulic cranking system shall consist of a hydraulic cranking motor, one or more piston-type accumulators with sufficient capacity for three consecutive cold engine starts without recharging, an oil reservoir, a hand-operated pump for charging the accumulator, and engine-driven pump for charging the accumulator, a pressure gage, and all the connecting lines, valves, and fittings to provide a complete cranking system. The hydraulic accumulators shall also provide power to the hydraulic motor driven pump in the drain down system (see 3.6.12).

3.6.7 Filter-separator. The filter-separator shall meet all requirements of MIL-F-8901 and other requirements specified herein. The filter-separator rated capacity shall not be less than 600 gallon per minute (gpm). The filter-separator shell, head, interior compartments, and supports shall be of aluminum, and shall be hydrostatically tested for a pressure of not less than 225 pounds-force per square inch gage (psig). The pressure vessel shall be designed and fabricated in accordance with ASME Boiler and Pressure Vessel Code, Section VIII. No leakage shall occur between compartments. Supports and bracing for canisters and filters elements shall be adequate to prevent displacement of or damage to the elements, canisters, or seals due to jolting or vibration from transporting the unit over rough terrain or in aircraft. Fuel flow passages through the filter-separator shall be adequate to produce low flow velocities at maximum rated capacity and prevent carryover of free water droplets. The filter-separator shall incorporate a water sump with heat exchanger heated by the engine coolant to prevent ice formation; and shall include other components and sensing devices specified herein and required to meet performance requirments.

3.6.7.1 Automatic water drain. An automatic water drain with sight glass and auxiliary manual valve shall be provided to drain the water sump through a drain hose to atmosphere below the chassis. Discharge of fuel from the filter-separator shall be automatically stopped by indirect action of a sensing device when water in the sump reaches a pre-determined level.

3.6.7.2 Air eliminator and pressure relief valve. An automatic air eliminator shall be provided above the liquid line of the filter-separator. The air eliminator shall prevent moist fuel vapor from being exhausted to the

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atmosphere, and shall scavenge liquid fuel back into the system. A pressure relief valve, meeting the requirements of the ASME Boiler and Pressure Vessel Code, Section VIII, shall be provided in conjunction with either the air eliminator or the automatic water drain assembly so as to relieve overpressure within the filter-separator.

3.6.7.3 Canister. The filter-separator shall contain 30 Military standard conisters conforming to Drawings D13216E2773 and D13216E2774. The canister shall strip out suspended water drops on the inner surface of the polytetrafluoroethylene coated monel screen. Six spare canisters shall be supplied with each unit (see 3.6.4).

3.6.7.4 Filter elements. Each filter-separator shall contain 30 Military standard filter elements conforming to MIL-F-52308. In addition, 60 filter elements shall be furnished with each fuel servicing unit (see 3.6.4).

3.6.7.5 Tubing vents and drain connections. The aluminum or copper tubing used to vent the pressure relief valve, the air eliminator valve, the automatic water drain valve, and the manual water drain valve shall all extend down to within 12 inches ±1 inch of the bottom of the frame. The pressure relief and air eliminator lines shall be secured to the filter-separator. Both the manual and the automatic water drains shall terminate with 3/4-inch female hose connections. The lower compartment drain valve shall terminate in a 3/4-inch female hose connection.

3.6.7.6 Cover, lifting, and swing mechanism. A cover, lifting, and swing mechanism of the cam lever operated type shall be provided to permit easy filter element replacement. The lifting eyes shall be of sufficient strength to enable lifting the entire filter-separator unit.

3.6.7.7 Cover bolts and gaskets. The filter-separator shall have stainless steel swing bolts and nuts to fasten down the cover. The vessel flange shall be grooved to receive an o-ring type gasket.

3.6.8 Contamination monitor. A full flow contamination monitor conforming to MIL-M-81380, as modified by Figure 3 and other requirements herein, shall be provided downstream and outside of the filter-separator. The monitor housing shall contain one set of 40 monitor elements, and shall have a quick-opening end with easy accessibility for rapid replacement of the monitor elements. The quick-opening end shall be of the captive C-clamp, swing bolt, or similar design. A spare set of 40 monitor elements, mounted in a spare rack to fit the monitor housing, and an unmounted spare boxed set shall be furnished with each fuel servicing unit (see 3.6.4). Figure 3 is included for illustration purposes and is not intended to exclude other monitor case designs which satisfy the requirements specified herein.

3.6.9 Meter. A positive displacement, rotary type meter with reset register reading to 99,999 gallons and a recording register reading to 9,999,999 gallons shall be installed downstream from the filter. Both registers shall record in 1-gallon increments. The accuracy of the meter shall meet requirements of NBS H44. The material of the case, adaptor fittings, and covers shall be anodized aluminum alloy 356-T6. Bolts, nuts, and fasteners shall be stainless steel or cadmium-plated steel. Internal parts which may contact the metered fluid shall be stainless steel or aluminum. The recorder shall be drip proof and not affected by salt spray or atmosphere exposures, and shall be readable from the operator's position.

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3.6.9.1 Face. The register shall have a transparent, colorless, antistatic, plastic face. The face shall be of sufficient size to provide a full view of all digits.

3.6.9.2 Indexing. The register shall be a type that can be indexed throughout a full revolution in increments of 45° or less, to permit positioning of the register in any direction with relation to the meter housing.

3.6.9.3 Static pressure. The meter shall be capable of operating at a working pressure of 150 psi.

3.6.9.4 Flow pressure drop. The maximum pressure drop between inlet and outlet shall not exceed 3 psi when operating at rated capacity at 15° centigrade.

3.6.9.5 Meter error. The meter error shall not exceed 0.1 percent for any one predetermined flow rate and accuracy setting. The maximum error shall not exceed 0.3 percent for any one given accuracy adjustment at any flow rate ranging from 30 to 600 gpm.

3.6.9.6 Overload operation. The meter shall be capable of overload operation at 115 percent of maximum rated capacity without damage to the mechanism.

3.6.10 Valves and flow control. Valving shall be provided to permit accomplishment of all fueling, defueling, purging, bypassing, and automatic and manual control operations. The automatic hydraulically actuated flow control valve, with pilots and sensing devices, shall be provided and shall meet performance requirements specified herein. Aluminum manual gate valves shall be provided at each inlet and discharge hose connection, and butterfly type or fuel-defuel type valves shall be used for fueling-defueling routing. All manual valves shall be grouped at the operator's position and shall be readily accessible. A check valve shall be provided to positively prevent impure fuel from being introduced downstream of the filter-separator. A petcock shall be provided downstream of the contamination monitor for test sampling of the filtered fuel. Under the operating conditions specified herein, fuel or contaminants shall not bleed or leak through the valving to the lower pressure side when fueling, defueling, or operating from a pressure hydrant source.

3.6.11 Line strainer. A 100 mesh line strainer shall be provided in the inlet line to the pump. The strainer housing shall be of aluminum alloy with 6-inch flanged type pipe connections with grooved adapter fittings and removable cover. The strainer insert shall be of monel wire or stainless steel and shall be removable for cleaning.

3.6.12 Drain down system. A drain down system shall be provided to drain fuel from the various components of the fuel filtering system and from the low spots of the pipe connections. Check valves shall be provided to prevent cross connections or bypassing of contaminated fuel during fueling conditions. The drain down system shall be interconnected and shall have a hydraulic motor driven pump, with outlet hose connection and not less than 20 feet of hose, with connections, suitable for the intended use. Drain down, to

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approximately empty condition, shall be accomplished in not more than 10 minutes.

3.6.13 Piping, hose, and connections. Components shall be positioned for optimum piping arrangements, short connections, and unrestricted flow. Mitered pipe bends shall not be used, and flexible pipe connections of resilient material shall be used to isolate pump and engine vibrations from components rigidly mounted to chassis frame. Reinforced, nylon lined, synthetic rubber, or plastic jacketed hose shall be used for hydraulic connections, and size variations shall be limited. Metal tubing shall not be used between components, and when used on a component, such as the hydraulic actuated fuel control valve, must be heavy duty and protected from damage wherever possible. A Military standard 2-1/2 inch size center point nozzle receptacle and a 4-inch recirculation hose connection shall be provided on the inlet (suction) of the fuel servicing unit. The recirculation hose connection and the two suction hose connections shall comply with MS27027, Part No. MS27027-17, size 4-inch, with gasket conforming to MS27030, Part No. MS27030-9. Discharge hose connections shall comply with MS27023, the 4-inch size shall be Part No. MS27023-17 and the 2-inch sizes shall be Part No. MS27023-11. All loose caps, plugs, or pins shall have stainless steel attachments wires or brass chains to prevent loss.

3.6.14 Instrumentation. All gages, instruments, starting valves, and other controls shall be mounted in a common panel with protecting hinged cover. The panel shall be at the operator's position and shall be at proper height for easy reading and use. Hydraulic lines shall be grouped, tied, and supported to prevent chafing damage. Direct reading pressure differential gages with a calibrated range of 0-25 psig, 4-1/2 inch diameter and minimum operating pressure rating of 250 psig showing pressure drop across the filter-separator and across the contamination monitor, shall be provided; also a gage indicating lb of pressure at the discharge piping manifold and a vacuum gage calibrated in inches of mercury at the suction shall be provided on the instrument panel.

3.6.15 Electrical static grounding. All components of the unit shall be adequately bonded to the frame to dissipate static electrical charges to ground. Two grounding reels, each with 100 feet of No. 12 American Standard Wire Gage, stranded copper wire, or 1/8-inch diameter, performed, seven strands of 19 wires, steel cable, clear plastic covered, shall be provided. The free end of the cable shall be provided with a type PCL, item No. 1, 100-ampere clamp conforming to W-C-440. The reels shall be mechanically and electrically attached to the trailer. The electrical continuity from the cable to the body of the reel shall be by a minimum of three parts. The reels shall be adjacent and located near each end of frame. The resistance between the clamp reel, and trailer frame shall not exceed of 0.5 ohmmeter. The cable shall be contained on a spring loaded, interrupted gear, ratchet type reel. A cable stop, which will prevent damage to the clamps because of rapid rewind, shall be provided. Two copper clad steel ground rods, 6 feet long by 5/8-inch diameter, shall be furnished and stored on the unit (see 3.6.4).

3.6.16 Spare parts and tools. In addition to any special tools needed for operation or calibration of any part or component, the following spare parts shall be furnished and stowed (see 3.6.4) with each unit:

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a. One set of belts for each belt driven operation.
b. Four sets each of o-ring seals for the line strainer, filter-separator shell cover, and contamination monitor cover.

3.7 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. The color of the finish coat shall be specified (see 6.2).

3.8 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be "FUEL SERVICING UNIT, AIRCRAFT, 600 GPM, AIR TRANSPORTABLE".

3.9 Instruction plates. Each fuel servicing unit shall be equipped with instruction plates, including warning and caution, describing important procedures to be followed in operating, servicing, and maintenance, particularly for the engine, filter-separator, automatic flow control valve, pressure regulator, and fuel-defuel valving. Plates shall be of a durable material which will provide long life legibility, and shall be securely affixed at appropriate locations providing good visibility. A caution sign in red letters not less than 2-inches high reading "NO SMOKING WITHIN 50 FEET" shall be stenciled in a prominent location. Each manual fueling or defueling valve described in he operating instructions shall be permanently and distinctively identified by a suitable method, such as securely wiring an inscribed or embossed metal tag to the valve, or an embossed metal bead by brazing. Color symbols or directional arrows with the words "FUEL" and "DEFUEL" shall be provided.

3.10 Filter-separator element and fuel contamination monitor change criteria data plates. An element change criteria data plate shall be mounted adjacent to the filter-separator pressure differential gage. The data plate shall be entitled "FILTER-SEPARATOR ELEMENT CHANGE CRITERIA". The plate shall indicate that the elements shall be changed at 15 psi pressure differential, or after 1 year's service, whichever occurs first. A second data plate, similar to above, shall be mounted adjacent to a differential pressure gage to be provided and used for the fuel quality monitor. The data plate shall be entitled "FUEL CONTAMINAION MONITOR CHANGE CRITERIA". The plate shall indicate that the elements shall be changed at 15 psi pressure differential or when the unit triggers (shuts off the flow), whichever occurs first.

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3.11 Workmanship.

3.11.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall bedone by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on fuel servicing unit when a first article is required (see 3.2). This inspection shall include the examination of 4.3 and the tests of 4.4.1 through 4.4.1.8.2. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each fuel servicing unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 First article tests. The first article fuel servicing unit shall be subjected to the tests specified in 4.4.1.1 through 4.4.1.8.2. Failure to pass one or more tests shall be cause for injection.

4.4.1.1 Mobility tests. The fuel servicing unit, in a drained condition, shall be towed by a truck tractor or a vehicle having a pintle height of 10 to 30 inches over paved highway at an average speed of 45 miles per hour (mph) for a distince of 35 miles, over gravel roads at an average speed of 25 mph for a distance of 15 miles, and over Belgian block or equivalent at an average speed of 10 mph for a distance of 35 miles. At least 10 short-radius right turns, 10 short-raduis left turns, and five sudden stops shall be conducted. Throughout the test, the fuel servicing unit shall be observed as to its trackability, ease of handling, tendency to yaw, and tendency to sway. Upon completion of the tests, the unit shall be inspected for misalinement, distortion, or other damage sustained as a result of the testing. Undesirable roadability features and any damage resulting from the test shall be cause for rejection. The causes for damage or malfunction shall be determined, and the unit shall be corrected and retested until proper mobility performance is attained.

4.4.1.2 Filter-separator tests. The filter-separator, complete with canisters, elements, automatic water discharge, automatic air eliminator, and automatic flow control and pressure regulating valves, shll be laboratory tested to determine conformance to the requirements of MIL-F-8901 and this specification. The test fluid shall be jet fuel JP-5.

4.4.1.3 Operational tests. With all elements and canisters installed in he filter-separator and all elements installed in the contamination monitor, the completely assembled fuel servicing unit shall be subjected to an operational

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test of approximately 5 hours duration. Suitable unpressurized sources of JP-5 jet fuel shall be provided. Two 4-inch suction hoses shall be connected between the fuel source being used and the inlet connections of the fuel servicing unit. Two 2-inch dispensing hoses, each 75 feet long, with 2-inch overwing nozzles shall be connected to each of the 2-inch discharge connections of the fuel servicing unit. A 4-inch hose, 75 feet long, with 2-1/2 inch center point nozzle shall be connected to the 4-inch discharge connection of the fuel servicing unit. The diesel engine shall be provided with JP-5 jet fuel for the operational test. Failure of the fuel servicing unit or any component part or accessory to perform or function properly or to meet the requirements specified herein shall be cause for rejection.

4.4.1.3.1 First and third hour fueling test. Using JP-5 test fuel source, a 50 psi pressure ±5 psi, shall be maintained at the discharge piping manifold; and a controlled vacuum of the not less than 8 inches and not more than 10 inches of mercury shall be developed at the suction connection while discharging to atmosphere through 75 feet of 4-inch diameter fuel hose through the center point nozzle at flow rates of 300 and 600 gpm. A water separation test shall be made during the third hour fueling test as follows: Introduce 15 gallons of water into the fuel source or suction inlet over a period of 15 minutes. While pumping the contaminated fuel through the fuel servicing unit, the analysis of water content shall be in accordance with the methods and tests in MIL-F-8901. Flow rates and pressures shall be determined accurately, and a pressure transducer recording of surge pressures at 1/2 second nozzle closure rate and 600 gpm flow shall be made. The performance of all components and controls, including the automatic water discharge device, shall be observed.

4.4.1.3.2 Second hour defueling test. Using the jet fuel source as a simulated aircraft with a 6-foot positive head, demonstrate ability to defuel through the 4-inch hose and 2-1/2 inch center point nozzle and discharge through the normal inlet (suction) connection of the fuel servicing unit to atmosphere, with center point hose valve open, approximately half open, without collapsing the 4-inch center point hose. The hose shall be designed to remain open without collapsing at a vacuum of 10 inches of mercury. Demonstrate ability to collapse the above hose and evacuate the fuel therefrom. Visually examine all components and controls for conformance to the requirements specified herein.

4.4.1.3.3 Fourth hour test. Using JP-5 as test fuel, demonstrate ability to develop a pressure of 30 psi at the discharge piping manifold, and a vacuum of not less than 8 inches in mercury at the suction connection, while discharging through two 75 foot lengths of 2-inch fuel hose to atmosphere at rates of 100 to 150 gpm through each of the two overwing nozzles.

4.4.1.3.4 No-flow tests. At the end of each hour, the unit shall be tested at no-flow condition. With no-flow on discharge side, and without operator's attention, the engine shall decelerate to idle speed, shall continue to operate at idling speed for 15 minutes without shutdown or other adverse effects, and shall automatically accelerate to proper pumping speed when nozzle is opened for next phase of tests. An accumulated total in excess of 20 minutes of other no-flow conditions because of malfunctions or excessive vibration or leaks, which cannot be stopped during the tests, shall be cause for rejection. A complete retest shall be accomplsihed after corrections are made.

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4.4.1.4 Environmental tests. The following environmental tests shall be conducted on the fuel servicing unit in accordance with the specified methods of MIL-STD-810.

4.4.1.4.1 High temperature test. The fuel servicing unit shall be subjected to high temperature in accordance with method 501.1, precedure I. No adverse effects that would affect operation at normal temperature shall result from this exposure. The chamber temperature shall be reduced to 125°, and the first hour test criteria of 4.4.1.3.1 shall be used. A duration of not less than 1 minute shall be maintained for each flow rate stipulated. The engine shall then be stopped for 5 minutes and restarted. The unit shall be operated at 600 gpm with 50 psi +5 psi at the discharge piping manifold with the single-point nozzle for 1/2 hour. The engine shall be stopped for 5 minutes and restarted. The engine shall be operated at idle speed with th single-point nozzle in the open position for 1/2 hour. All engine starts shall be accomplished within 5 minutes of the original attempt. The fuel servicing unit shall operate satisfactorily and perform its intended functions throughout this test. There shall be no evidence of leakage, damage, thermal degradation, or impending failure of any portion of the unit. The engine coolant temperature shall not exceed that recommended by the manufacturer of 10° below the boiling point rating of the unit, whichever is lower, and there shall be no loss of coolant. The engine shall be restarted without auxiliary cooling or electrical power.

4.4.1.4.2 Low temperature test. The fuel servicing unit shall be subjected to low temperature in accordance with method 502.1, procedure I, except after a -40°F soak, the chamber temperature shall be raised to -25°F and stabilized for 24 hours. The engine shall be started within 10 minutes of the initial attempt and shall demonstrate the ability to pump fuel at 300 gpm for a minimum period of 10 minutes. The fuel servicing unit shall operate satisfactorily and perform its intended function. The unit shall be inspected during the test and there shall be no evidence of leakage, damage, degradation, failure, or impending failure of any portion of the unit.

4.4.1.5 Drain down system test. The drain down system shall be operated discharging fuel into a separate container. The time required to remove the fuel shall be measured to determine conformance to 3.6.12.

4.4.1.6 Hydrostatic pressure test. All inlet and outlets of complete unit shall be plugged or capped with quick-discount fittings and the system pressurized with Stoddard solvent to 225 psig pressure. Any leaks, seal failures, or pressure drops shall be cause for rejection until such failure is corrected. Peening to close leaks is not acceptable. The air eliminator shall be plugged or removed and filtering elements removed during the test.

4.4.1.7 Ramp clearance. The fuel servicing unit, in a drained condition, shall enter, negotiate, and leave a ramp with an approach angle of 20° in both the forward and reverse directions. The length of the ramp, measured along the slope, shall be greater than the unit length. The bottom an top of the ramp shall end in horizontal landings which form a sharply defined 20° angle with the ramp.

4.4.1.8 Acceleration load tests. The following tests shall be performed to determine compliance with the acceleration load requirements specified in

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3.6.2.1, 3.6.2.2., and 3.6.3. If it can be shown by computation that the equipment and all components will conform to the requirements specified herein, the computation may be substituted for the test.

4.4.1.8.1 Flight and taxiing load tests. The equipment shall be attached in the manner specified for loading in an aircraft. The unit shall be slid or rolled down an inclined plane or otherwise accelerated and stopped in such a manner as to produce the accelerations specified in 3.6.2.1, as measured by an accelerometer mounted on the unit. Following these tests, the equipment shall be examined, operated, and subjected to the performance tests specified herein. Evidence of permanent deformation of structural members or faiure to operate and meet the performance requirements specified herein shall be cause for rejection. If application of the dynamic load to equipment along the vertical axis imposes undue complications on test equipment, the vertical acceleration may be applied statically.

4.4.1.8.2 Emergency landing load tests. Upon successful completion of the tests specified in 4.4.1.8.1, the equipment shall be subjected to the accelerations listed below in the order specified. Following these tests, the equipment shall be examined. Breaking loose of the equipment from the tiedown, or separation of a component from the main body of the equipment, shall be cause for rejection. The item need not be serviceable after being subjected to the following accelerations:

a. 4-1/2g as specified in 3.6.2.2(b).
b. 8g as specified in 3.6.2.2.(a).

If application of the dynamic load to equipment along the vertical axis imposes undue complications on test equipment, the vertical acceleration may be applied statically.

4.4.1.9 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample and shall be the last fuel servicing unit delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touchup or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.4.2 Quality conformance tests. Each production fuel servicing unit shall be tested for all operating functions and for adjustment and calibration of control devices, instruments, and safety devices, under actual fuel flow conditions. Each production fuel servicing unit shall be hydrostatically tested in accordance with 4.4.1.6. Failure to pass any phase of these tests shall because for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.6 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type I, method I.

5.1.1.7 Pumps. Interior surfaces of the pumps shall be coated with type P-10 preservative in a manner to insure thorough coating of all interior parts and surfaces. All openings into the pumps shall be sealed with caps, plugs, or waterproof tape.

5.1.1.8 Filter-separator. All openings into the filter-separator shall be sealed with caps, plugs, or waterproof tape.

5.1.1.9 Filter elements. Each spare filter element shall be preserved and packaged method IA. The submitted shall be at the contractor's option.

5.1.1.10 Hydraulic system. The hydraulic system shall be filled to the operating level with hydraulic fluid required for operation.

5.1.1.11 Radiator fronts and engine housings. When specified (see 6.2), radiator fronts and any large openings in the engine housings shall be covered with barrier material conforming to PPP-B-1055, class E-2 or M-1. The barrier material shll be secured with tape conforming to PPP-T-60, type IV. All

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vents, louvers, and other openings in the engine housings shall be sealed withtape specified herein.

5.1.1.12 Rubber tires. Tires shall be inflated to 10 lb above the specified operating pressure.

5.1.1.13 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.14 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.15 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts. When specified (see 6.2), the repair parts shall be preserved in accordance with level A requirements of MIL-R-196, or when parts are not specifically covered in MIL-R-196, an applicable submethod of preservation of MIL-P-116 shall be used.

5.1.2 Level C. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The fuel servicing units shall be shipped uncrated. Disassembled components and accessories for each fuel servicing unit shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type; or PPP-B-621, class 2. The contents sahll be cushioned, blocked, and braced to prevent movement within the containers. The repair parts and additional spare filter elements and canisters for each fuel servicing unit shall be packed separately in a container as specified herein. Where praticable, the disassembled components, accessories, repair parts, spare filter elements, and canisters shall be positioned and secured to the fuel servicing unit by bolting or with steel strapping in such a manner as not to interfere with towing or lifting the fuel servicing unit with slings.

5.2.2 Level C. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The trailer-mounted fuel servicing unit, designed for transferring and filtering light hydrocarbon fuels for direct aircraft fueling and defueling operations, is intended to be used at tactical airfields in conjunction with pillow tank type storage systems. The fuel servicing units may be utilized to augment a permanent airfield installation during normal fuel systems repair or as an auxiliary backup in case of accidental disruption of fuel delivery components.

6.2 Ordering data. Acquisition documents should specify the following:

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110. Components missing or not as specified (see 3.3).

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval (see 3.2).
c. Color of finish coat required (see 3.7).
d. When radiator fronts and engine housings require a covering (see 5.1.1.11).
e. Level of preservation and level of packing required (see 5.1 and 5.2).
f. When repair parts shall be preserved in accordance with level A of MIL-R-196 (see 5.1.1.15).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the ASPR. The first article should be a first production fuel servicing unit or it may be a standard production fuel servicing unit from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, tests, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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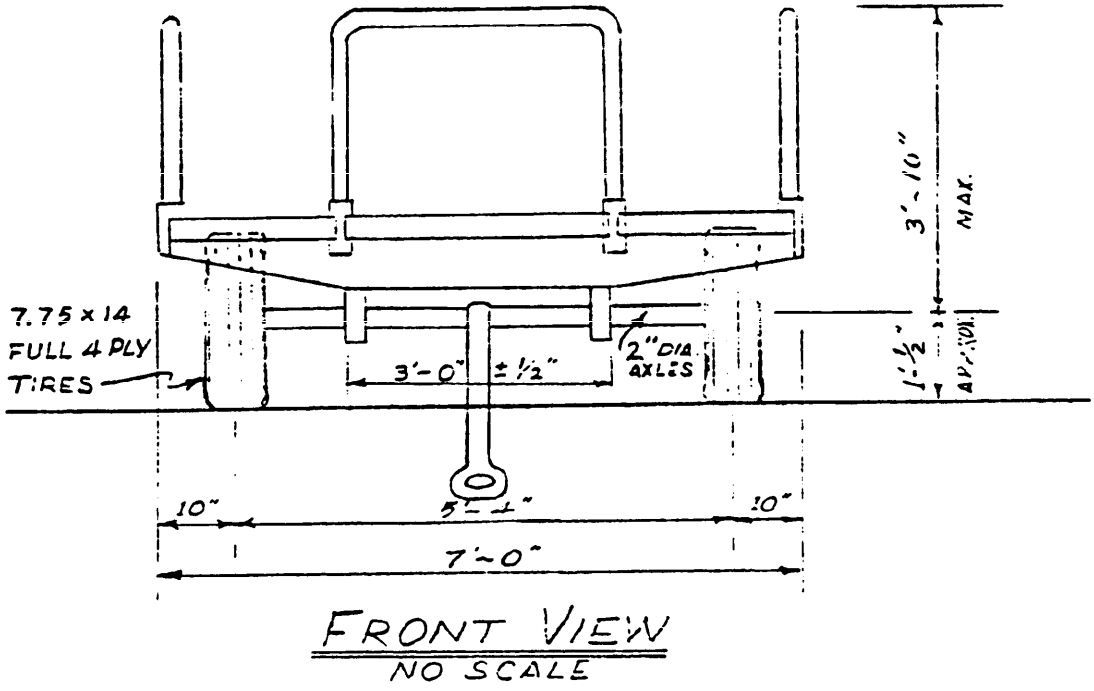
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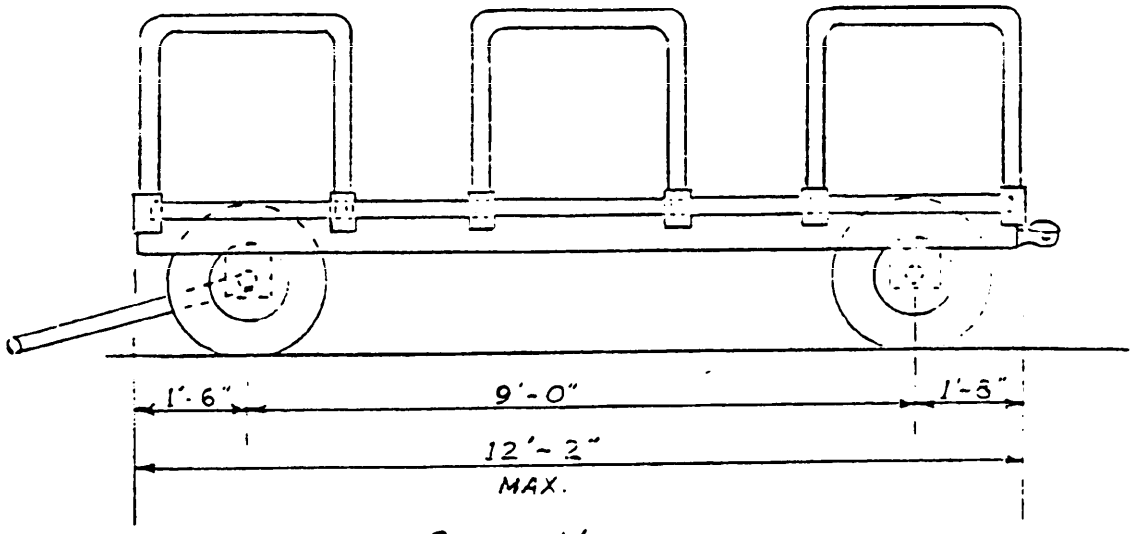
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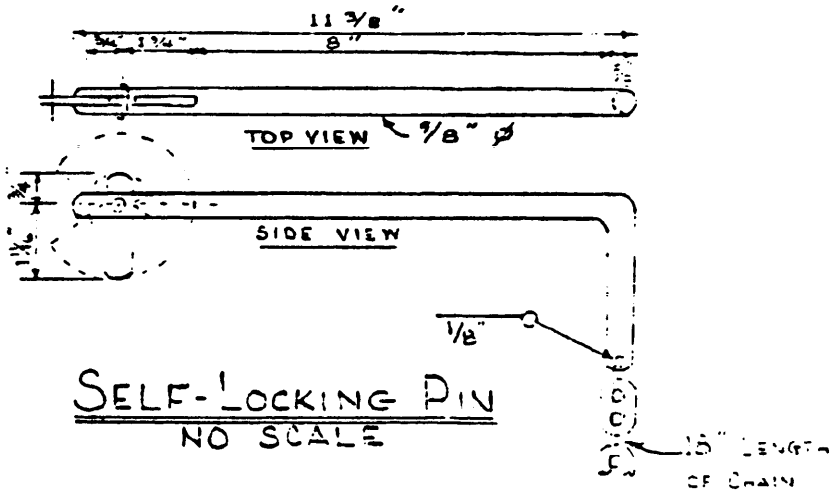
FRONT VIEW
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SIDE VIEW
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Figure 1. Trailer, transporter.

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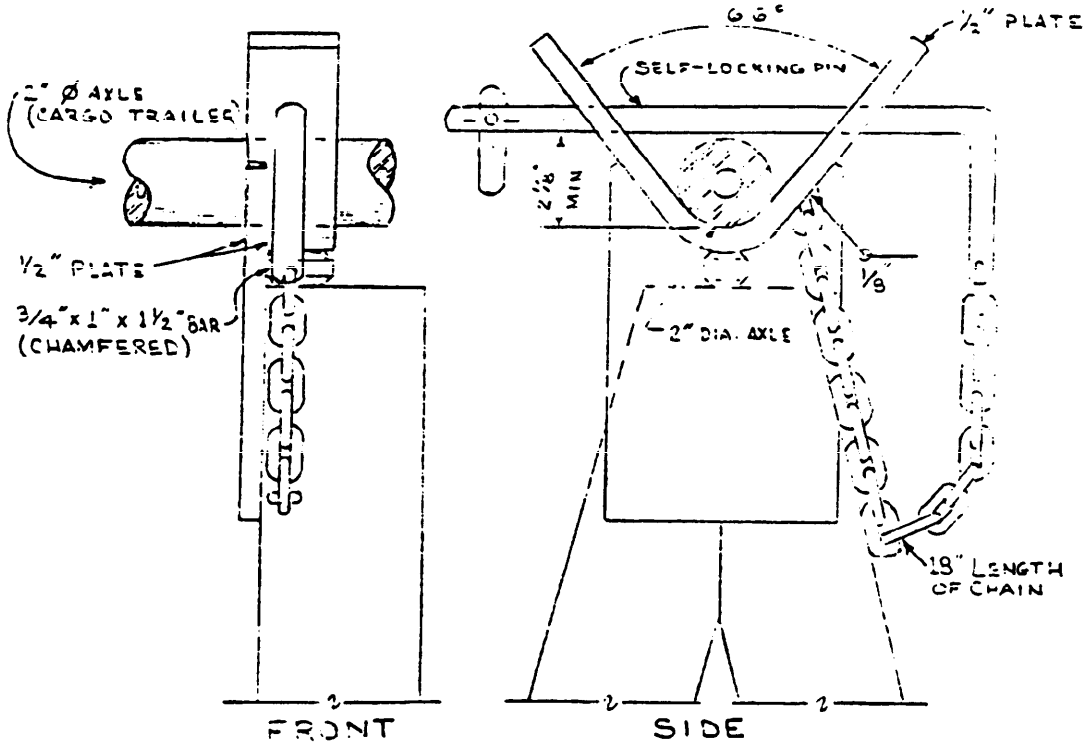


Figure 2. Cargo trailer support.

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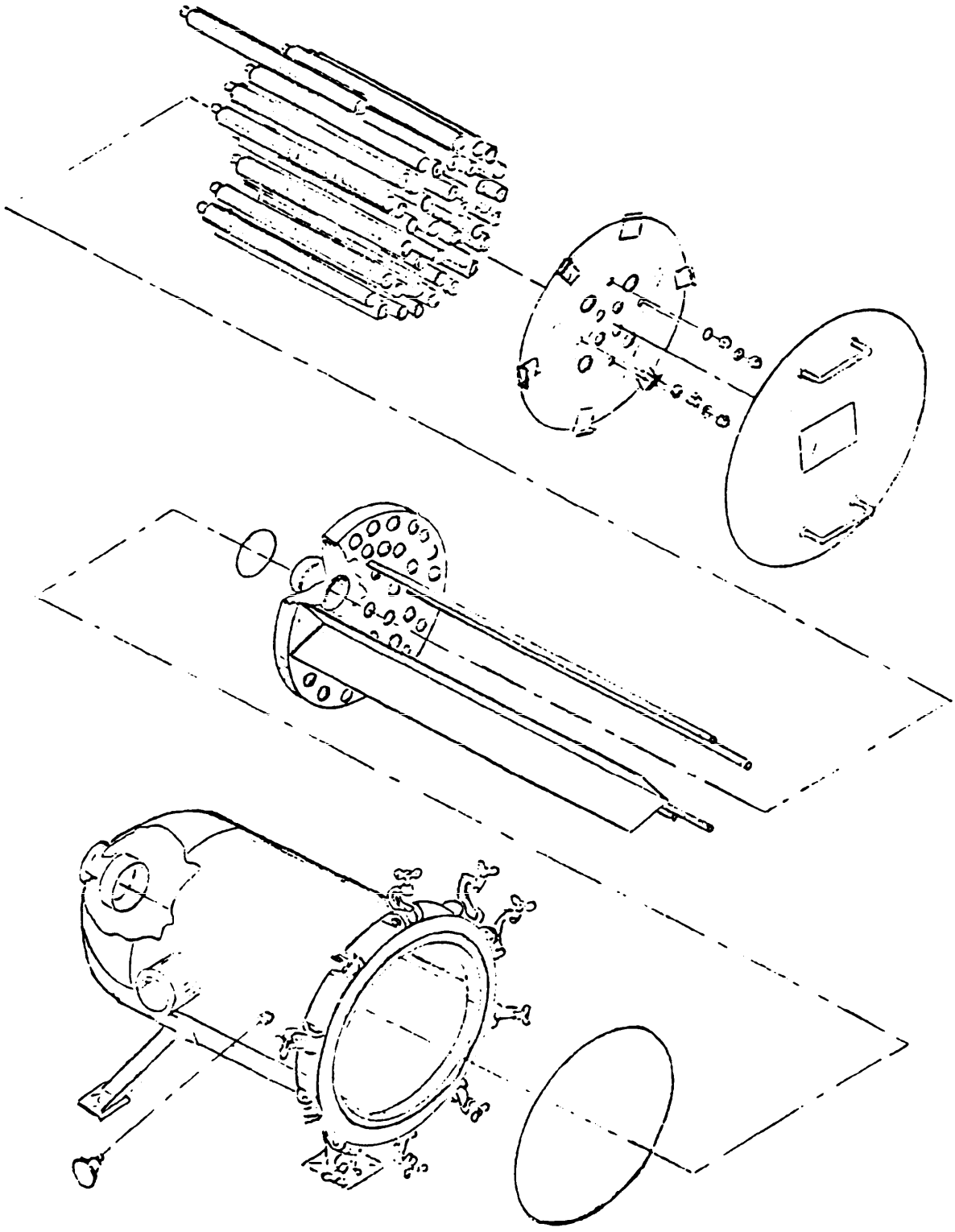


Figure 3. Contamination monitor.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
L-F-28570B(YD)	FUEL SERVICING UNIT, AIRCRAFT, TRAILER MOUNTED, 600 GPM, AIR TRANSPORTABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

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